Sorting

Merge Sort

- Merge sort is based divide and conquer strategy.
- Can be implemented as external sorting, when the dataset is so big that it is impossible to load the whole dataset into memory, here sorting is done in chunks.
- Merging is the process of combining two sorted files to make one bigger sorted file.
- Selection is the process of dividing a file into two parts: k smallest elements and n – k largest elements.
- Selection and merging are opposite operations
  - selection splits a list into two lists
  - merging joins two lists to make one list

Algorithm – Merge Sort

1. **Divide:** Divide the list or array recursively into two halves until it can no more be divided.
2. **Conquer:** Each subarray is sorted individually using the merge sort algorithm.
3. **Merge:** The sorted subarrays are merged back together in sorted order. The process continues until all elements from both subarrays have been merged.

Index	0	1	2	3	4	5	6
Values	9	4	7	6	3	1	5

1

0	1	2	3
9	4	7	6

4	5	6
3	1	5

2

0	1
9	4

2	3
7	6

12

4	5
3	1

6
5

3

0
9

1
4

7

2
7

3
6

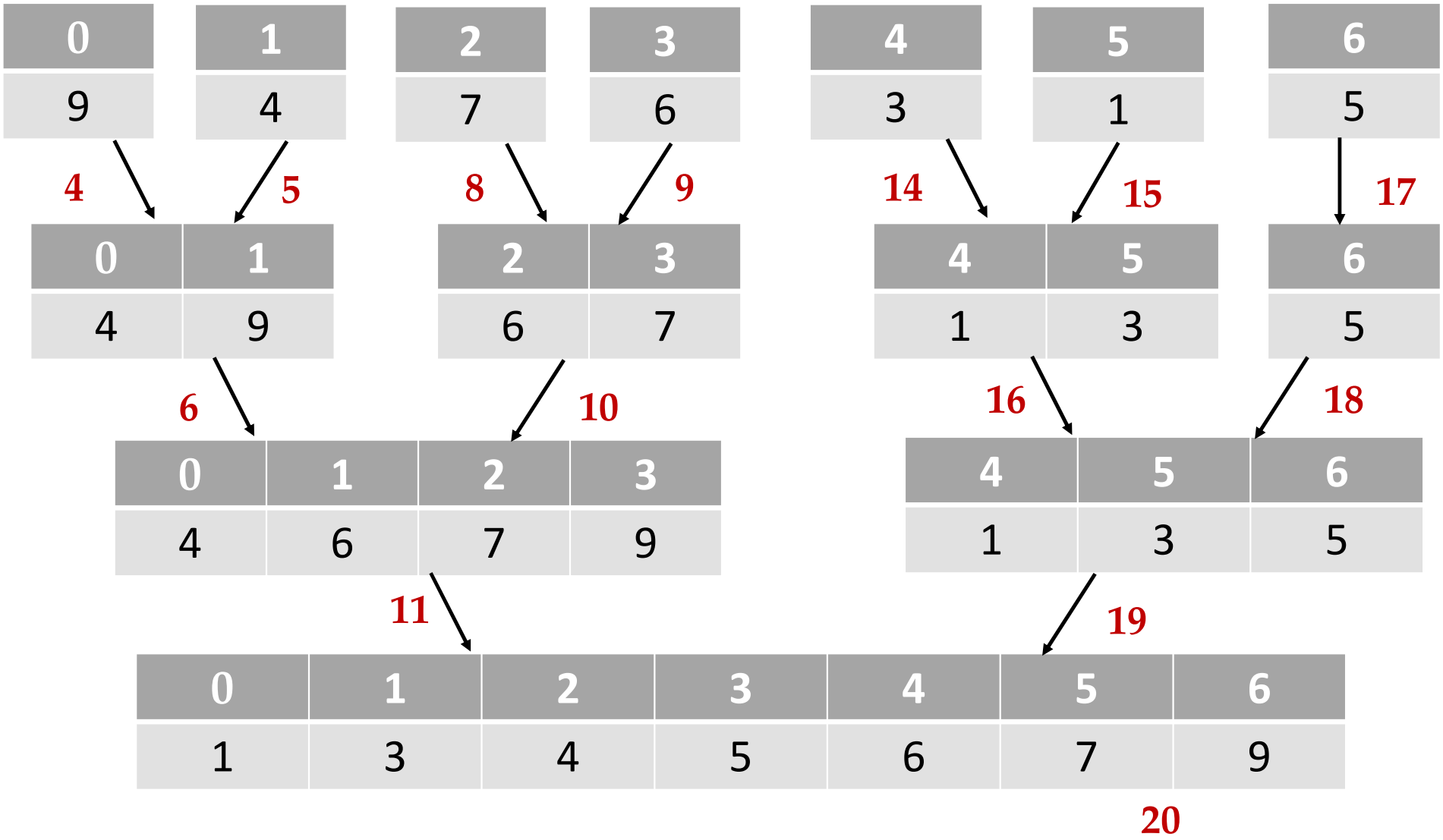
13

4
3

5
1

17

6
5


0 1 2 3 4 5 6
9 4 7 6 3 1 5
4 5 8 9 14 15 17
0 1 2 3 4 5 6
4 9 6 7 1 3 5
6 10 16 18
0 1 2 3 4 5 6
4 6 7 9 1 3 5
11 19
0 1 2 3 4 5 6
1 3 4 5 6 7 9
20

Performance – Merge Sort

- Worst case complexity : O(nlogn)
- Best case complexity : O(nlogn)
- Average case complexity : O(nlogn)
- Worst case space complexity: O(n) auxiliary

Quick Sort

- It is an example of a divide-and-conquer algorithmic technique.
- It is also called partition exchange sort.
- It uses recursive calls for sorting the elements, and it is one of the famous algorithms among comparison-based sorting algorithms.
- Divide: The array A[low ...high] is partitioned into two non-empty sub arrays A[low ...q] and A[q + 1... high], such that each element of A[low ... high] is less than or equal to each element of A[q + 1... high]. The index q is computed as part of this partitioning procedure.
- Conquer: The two sub arrays A[low ...q] and A[q + 1 ...high] are sorted by recursive calls to Quick sort.

Algorithm – Quick Sort

The recursive algorithm consists of four steps:

- If there are one or no elements in the array to be sorted, return.
- Pick an element in the array to serve as the "pivot" point. (Usually the left-most element in the array is used.)
- Split the array into two parts – one with elements larger than the pivot and the other with elements smaller than the pivot.
- Recursively repeat the algorithm for both halves of the original array.

Tip: In this algorithm, we place the pivot element at right place.

0	1	2	3	4	5	6	7	8	9	10	11
48	44	19	59	72	80	42	65	82	8	95	68

```
public static int partition(int arr[], int low, int high)
{                                                  0        11
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 48
  left=low;            // left = 0
  right =high;         // right = 11

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;
```

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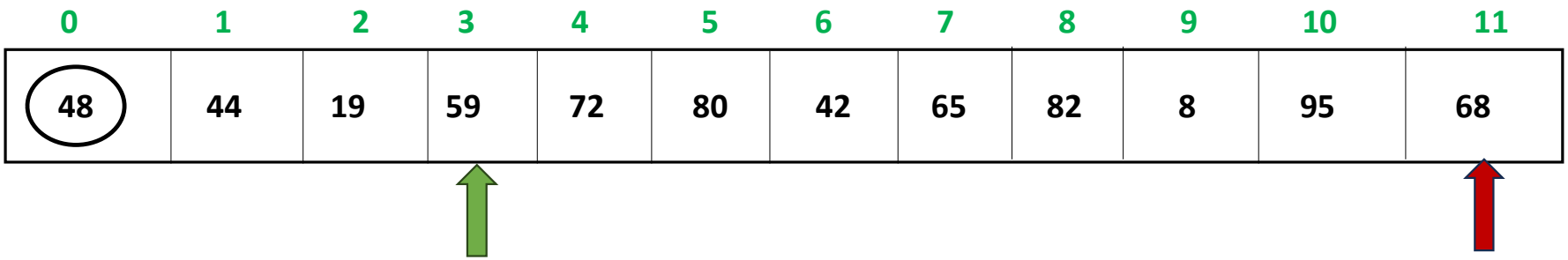


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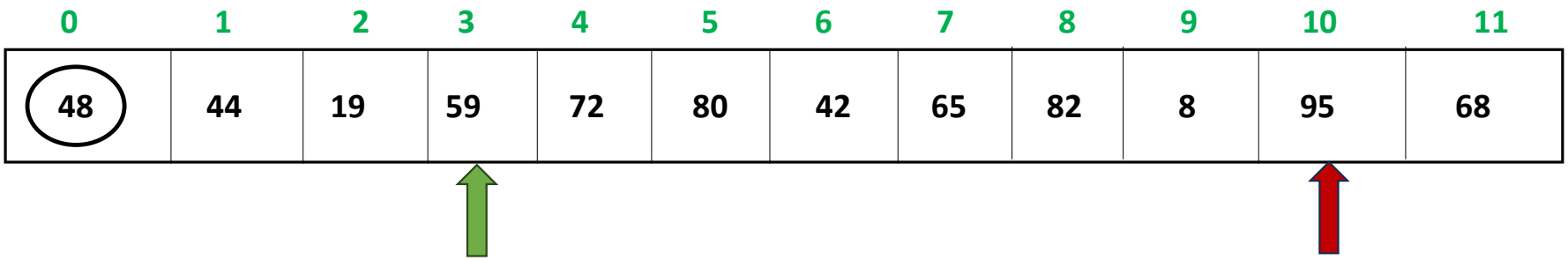


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  arr[low] = arr[right];
  arr[right] =pivot;

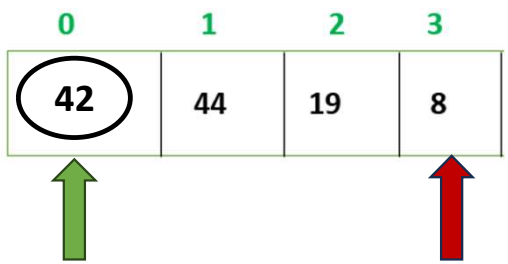
  return right;   // right = 4
}// end of function
```

```
public static void QuickSort(int arr[],
              int low,int high)
{                  //0       11
  int pivloc;

  if(low>=high)  return;

  pivloc = partition(arr,low,high); //4

  QuickSort(arr,low,pivloc-1);
              //    0        3
  QuickSort(arr,pivloc+1, high);
              //     5          11
}
```




```
public static int partition(int arr[], int low, int high)
{                                                  0       3
  int left, right, pivot, t;

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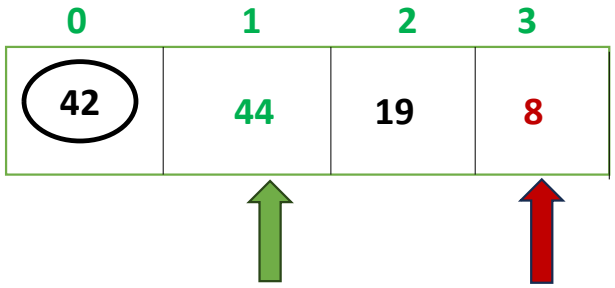


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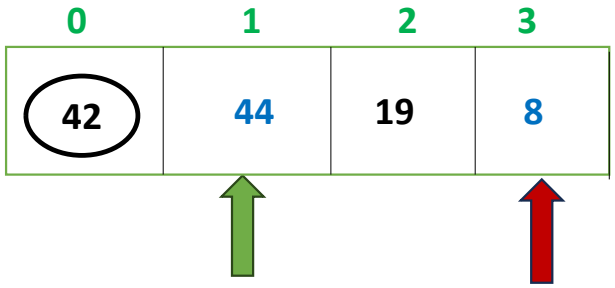


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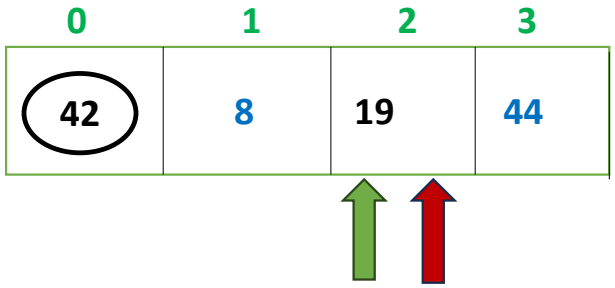


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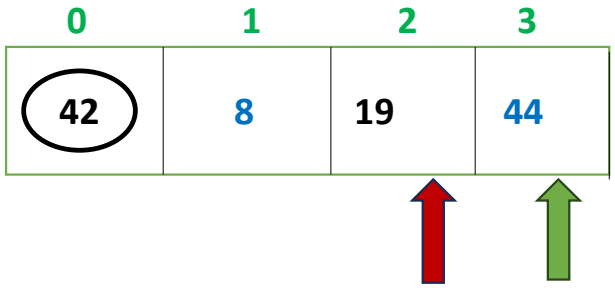


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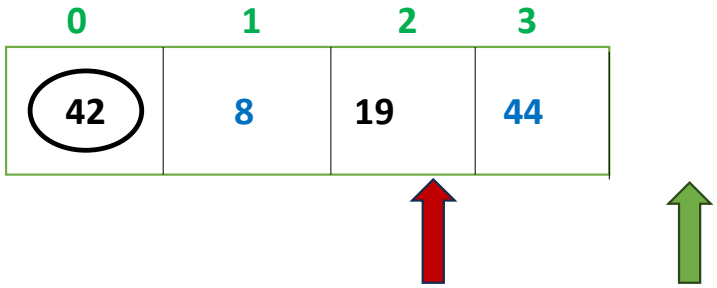


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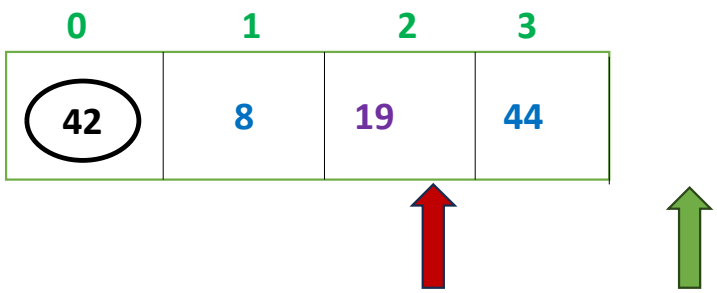
      right--;

   if(left < right)    //false
      {
        t=arr[left];
        arr[left] = arr[right] ;
        arr[right] = t;
        left++; right--;
      }
      else left++;
  }  // end of while loop
```

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     }
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  arr[low] = arr[right];
  arr[right] =pivot;

  return right;      // right = 2
}// end of function
```

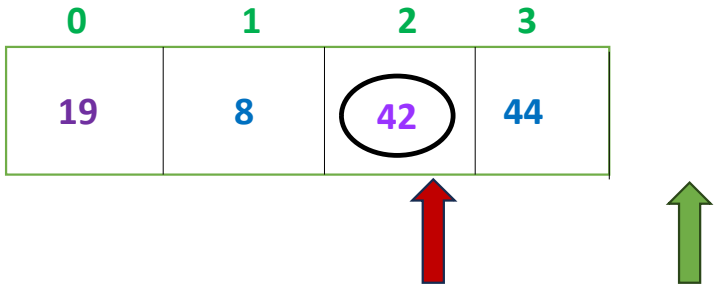


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}// end of function
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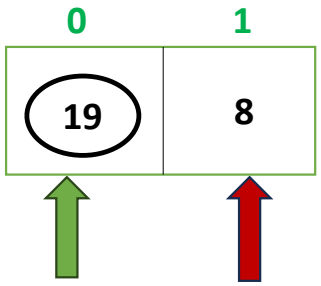
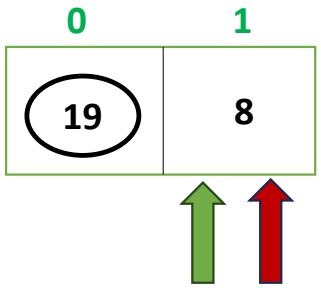
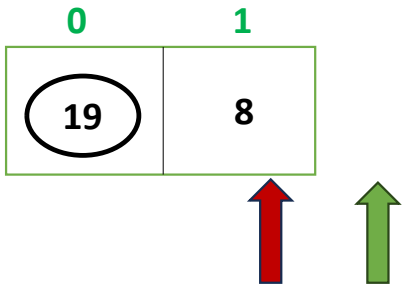
```
public static void QuickSort(int arr[],
              int low,int high)
{             //   0          3
  int pivloc;

  if(low>=high)  return;

  pivloc = partition(arr,low,high);  //2

  QuickSort(arr,low,pivloc-1);
              //     0        1
  QuickSort(arr,pivloc+1, high);
              //     3          3
}
```



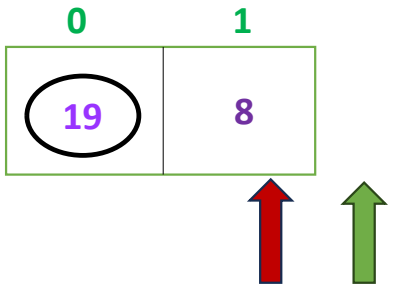

```
public static int partition(int arr[], int low, int high)
{                                                  0       1
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 19
  left=low;            // left = 0
  right =high;         // right = 1

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;
```



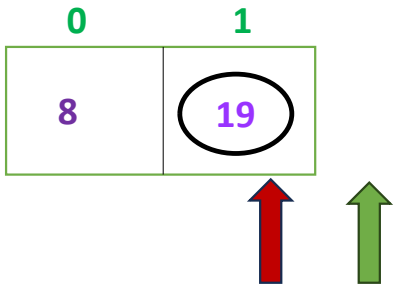

```
public static int partition(int arr[], int low, int high)
{                                                    0       1
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 19
  left=low;            // left = 0
  right =high;         // right = 1

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;
```




```
public static int partition(int arr[], int low, int high)
{                                                  0      1
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 19
  left=low;            // left = 0
  right =high;         // right = 1

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;

   if(left < right)    //false
     {
       t=arr[left];
       arr[left] = arr[right] ;
       arr[right] = t;
       left++; right--;
     }
     else left++;
  }  // end of while loop

  arr[low] = arr[right];
  arr[right] =pivot;

  return right;      // right =
1
}// end of function
```




```
public static int partition(int arr[], int low, int high)
{                                                  0      1
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 19
  left=low;            // left = 0
  right =high;         // right = 1

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;

   if(left < right)    //false
     {
       t=arr[left];
       arr[left] = arr[right] ;
       arr[right] = t;
       left++; right--;
     }
     else left++;
  }  // end of while loop

  arr[low] = arr[right];
  arr[right] =pivot;

  return right;      // right = 1
}// end of function
```



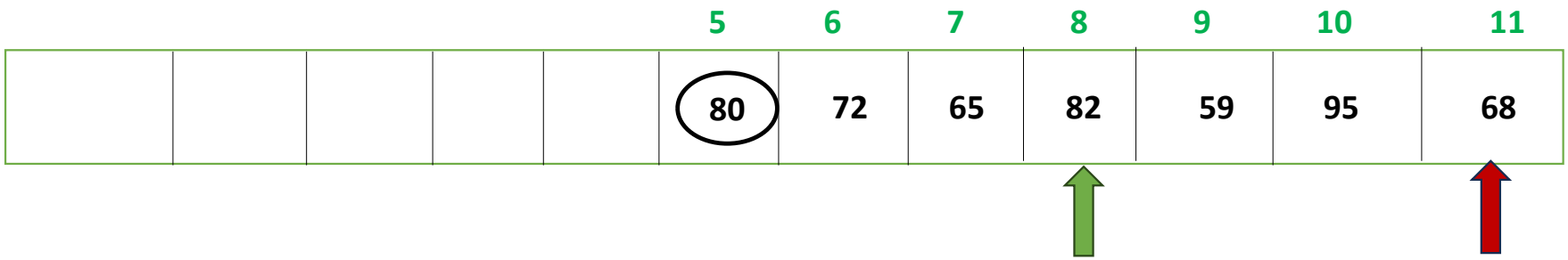

```
public static int partition(int arr[], int low, int high)
{                                                  0      1
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 19
  left=low;            // left = 0
  right =high;         // right = 1

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;

   if(left < right)    //false
      {
        t=arr[left];
        arr[left] = arr[right] ;
        arr[right] = t;
        left++; right--;
      }
      else left++;
   }  // end of while loop

   arr[low] = arr[right];
   arr[right] =pivot;

   return right;  // right = 1
}// end of function
```

```
public static void QuickSort(int arr[],
            int low,int high)
{
  int pivloc;

  if(low>=high)  return;

  pivloc = partition(arr,low,high);  //1

  QuickSort(arr,low,pivloc-1);
            //    0        0          //*****
  QuickSort(arr,pivloc+1, high);
            //    2        1          //*****
}
```

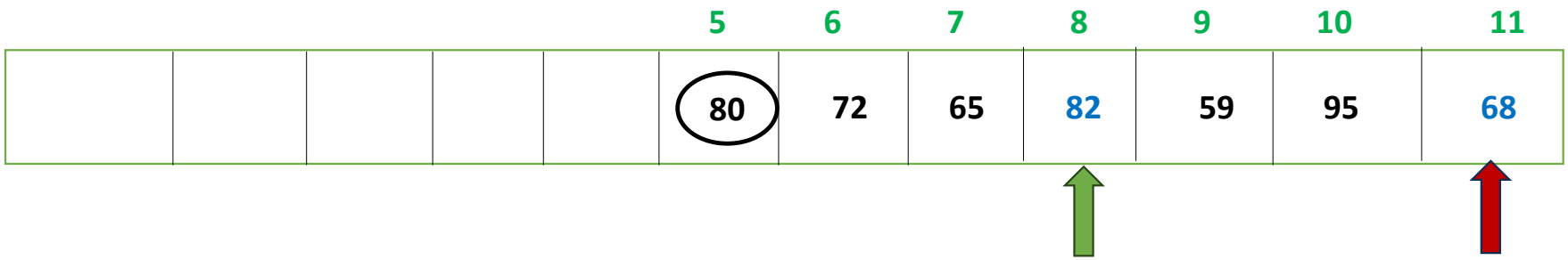
0	1	2	3	4	5	6	7	8	9	10	11
8	19	42	44	48	80	72	65	82	59	95	68

```
public static int partition(int arr[], int low, int high)
{                                                  5      11
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 80
  left=low;            // left = 5
  right =high;         // right = 11

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;

   if(left < right)
     {
       t=arr[left];
       arr[left] = arr[right] ;
       arr[right] = t;
       left++; right--;
     }
     else left++;
  }  // end of while loop

  arr[low] = arr[right];
  arr[right] =pivot;

  return right;   // right = 4
}// end of function
```

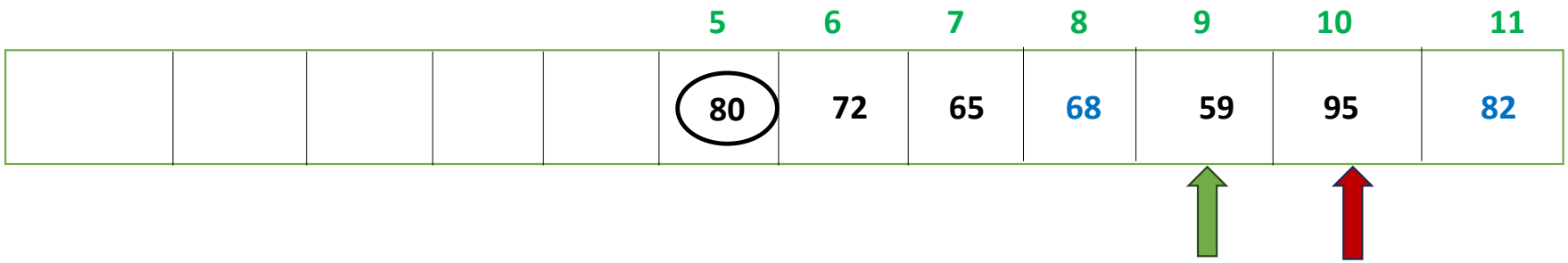
```
public static void QuickSort(int arr[],
            int low,int high)
{
  int pivloc;

  if(low>=high)  return;

  pivloc = partition(arr,low,high);

  QuickSort(arr,low,pivloc-1);
              //    0        3
  QuickSort(arr,pivloc+1, high);
              //    5        11
}
```



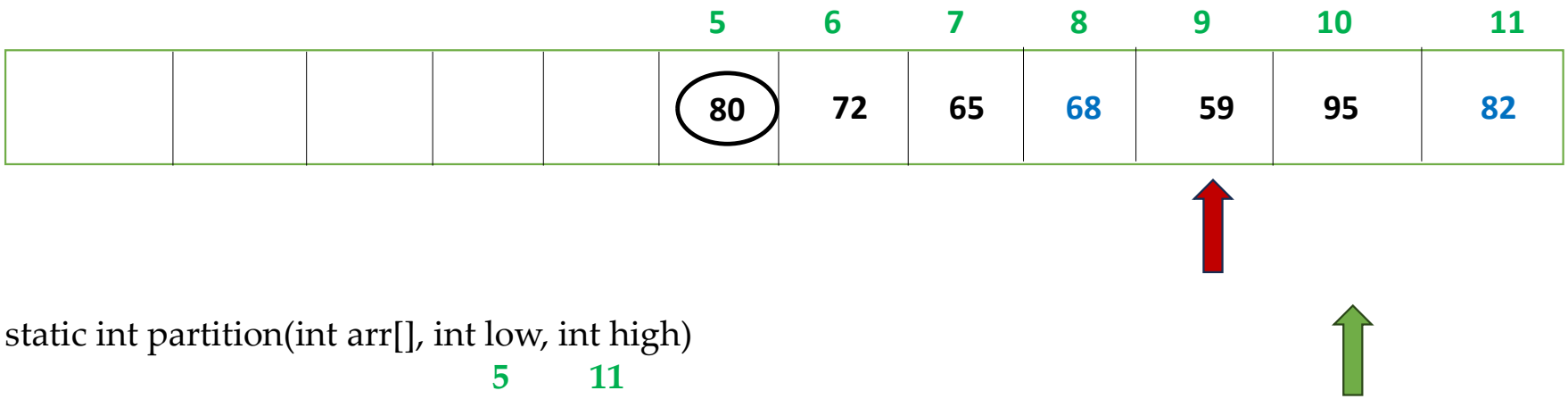

```
public static int partition(int arr[], int low, int high)
{                                                  5        11
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 80
  left=low;            // left = 5
  right =high;         // right = 11

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;
```



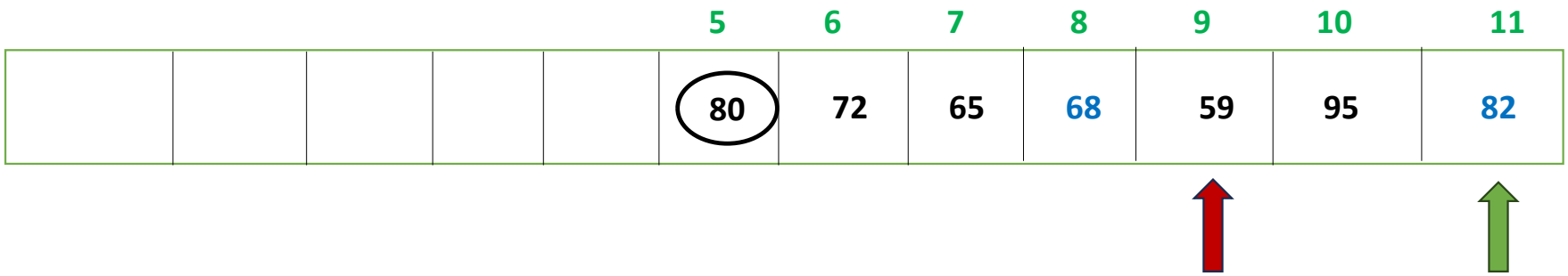

```
public static int partition(int arr[], int low, int high)
{                                                  5        11
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 80
  left=low;            // left = 5
  right =high;         // right = 11

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;

   if(left < right)    //true
     {
       t=arr[left];
       arr[left] = arr[right] ;
       arr[right] = t;
       left++; right--;
     }
     else left++;
   }  // end of while loop

   arr[low] = arr[right];
   arr[right] =pivot;

   return right;      // right =
2
}// end of function
```




```
public static int partition(int arr[], int low, int high)
{                                                  5        11
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 80
  left=low;            // left = 5
  right =high;         // right = 11

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;

   if(left < right)    //false
     {
       t=arr[left];
       arr[left] = arr[right] ;
       arr[right] = t;
       left++; right--;
     }
     else left++;
  }  // end of while loop
```




```
public static int partition(int arr[], int low, int high)
{                                                  5        11
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 80
  left=low;            // left = 5
  right =high;         // right = 11

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;
```

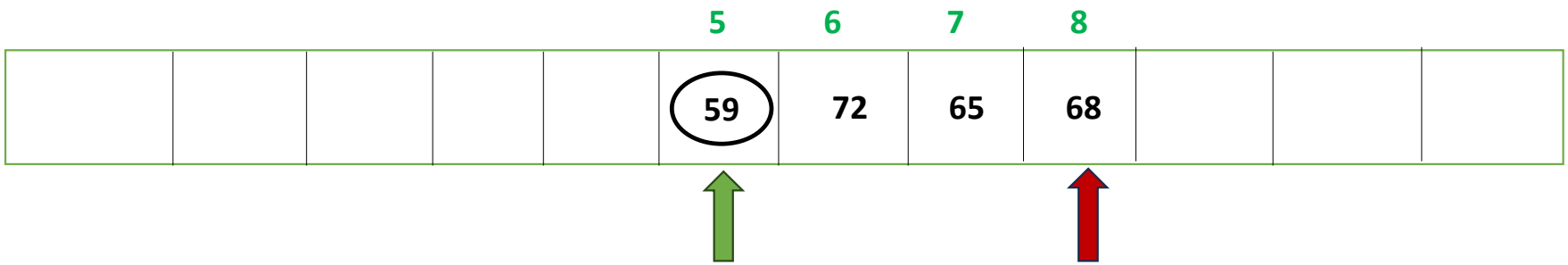



```
public static int partition(int arr[], int low, int high)
{                                                  5        11
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 80
  left=low;            // left = 5
  right =high;         // right = 11

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;

   if(left < right)    //false
     {
       t=arr[left];
       arr[left] = arr[right] ;
       arr[right] = t;
       left++; right--;
     }
     else left++;
   }  // end of while loop

   arr[low] = arr[right];
   arr[right] =pivot;

   return right;      // right =
9
}// end of function
```

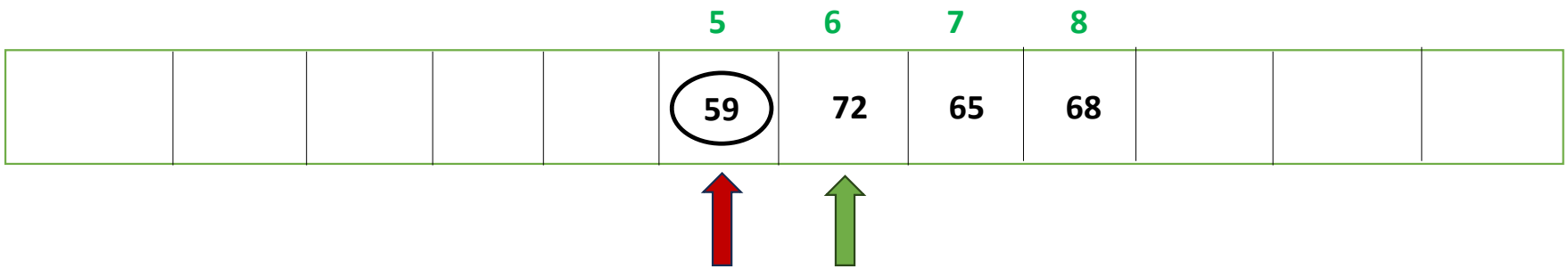
					5	6	7	8	9	10	11
					59	72	65	68	80	95	82

```
public static int partition(int arr[], int low, int high)
{                                                  5        11
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 80
  left=low;            // left = 5
  right =high;         // right = 11

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;

   if(left < right)    //false
     {
       t=arr[left];
       arr[left] = arr[right] ;
       arr[right] = t;
       left++; right--;
     }
     else left++;
   }  // end of while loop

   arr[low] = arr[right];
   arr[right] =pivot;

   return right;  // right = 9
}// end of function
```

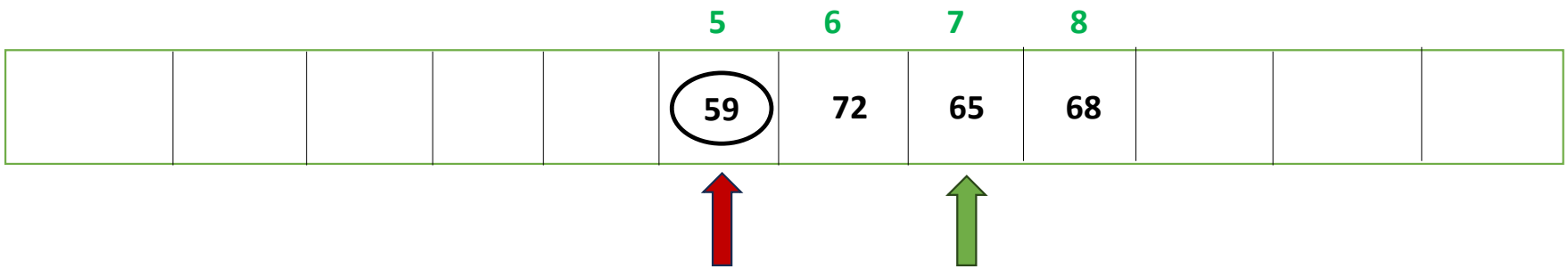
```
public static void QuickSort(int arr[],
              int low, int high)
{              // 5          11
  int pivloc;

  if(low>=high)  return;

  pivloc = partition(arr,low,high); //9

  QuickSort(arr,low,pivloc-1);
              //    5        8
  QuickSort(arr,pivloc+1, high);
              //     10        11
}
```



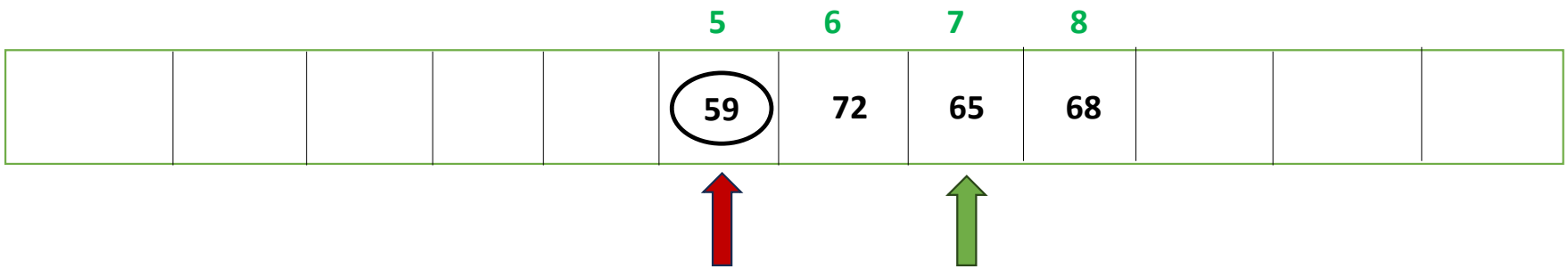

```
public static int partition(int arr[], int low, int high)
{                                                  5       8
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 59
  left=low;            // left = 5
  right =high;         // right = 8

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;
```




```
public static int partition(int arr[], int low, int high)
{                                                  5       8
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 59
  left=low;            // left = 5
  right =high;         // right = 8

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;
```

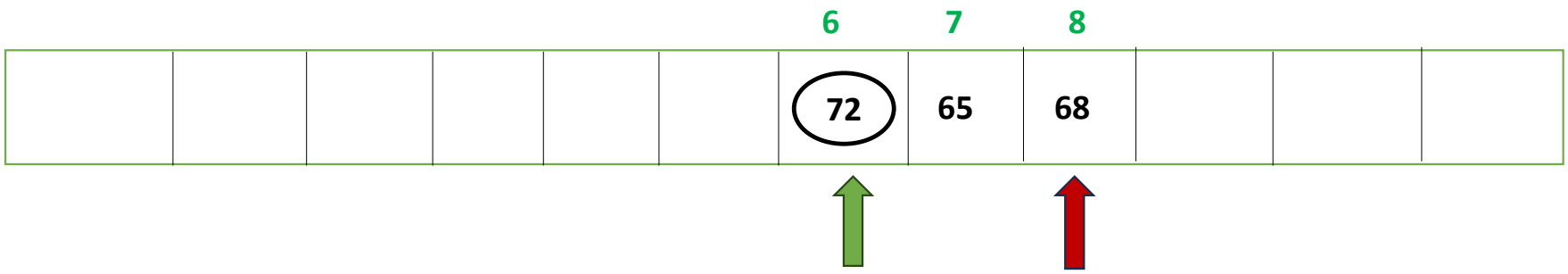



```
public static int partition(int arr[], int low, int high)
{                                                  5       8
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 59
  left=low;            // left = 5
  right =high;         // right = 8

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;

   if(left < right)    //false
     {
       t=arr[left];
       arr[left] = arr[right] ;
       arr[right] = t;
       left++; right--;
     }
     else left++;
   }  // end of while loop

   arr[low] = arr[right];
   arr[right] =pivot;

   return right;      // right = 5
}// end of function
```

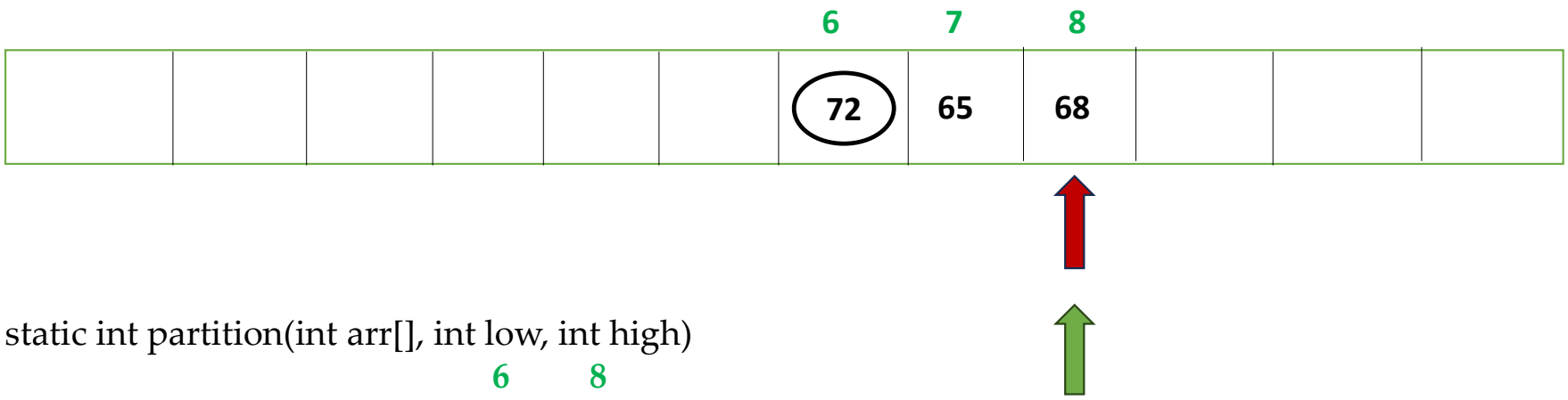



```
public static int partition(int arr[], int low, int high)
{                                                  5       8
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 59
  left=low;            // left = 5
  right =high;         // right = 8

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;

   if(left < right)    //false
     {
       t=arr[left];
       arr[left] = arr[right] ;
       arr[right] = t;
       left++; right--;
     }
     else left++;
   }  // end of while loop

   arr[low] = arr[right];
   arr[right] =pivot;

   return right;  // right = 5
}// end of function
```

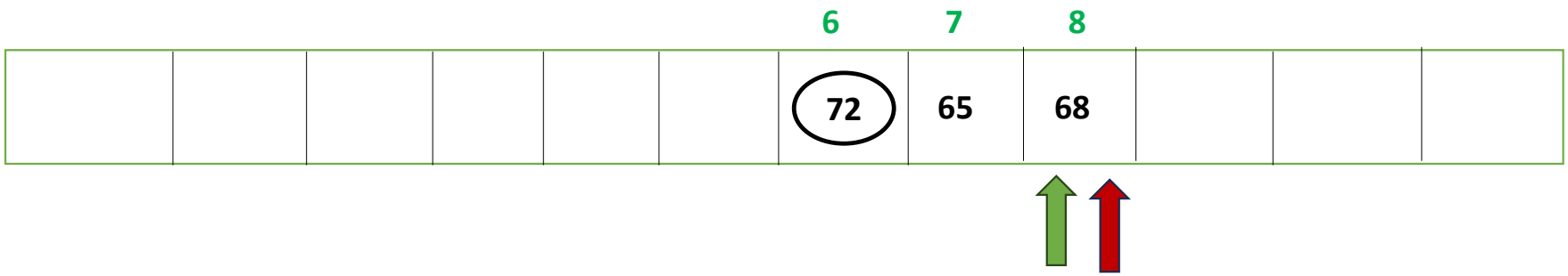
```
public static void QuickSort(int arr[],
              int low,int high)
{                  // 5        8
  int pivloc;

  if(low>=high)  return;

  pivloc = partition(arr,low,high); //5

  QuickSort(arr,low,pivloc-1);
                 //    5        4
  QuickSort(arr,pivloc+1, high);
                 //      6        8
}
```



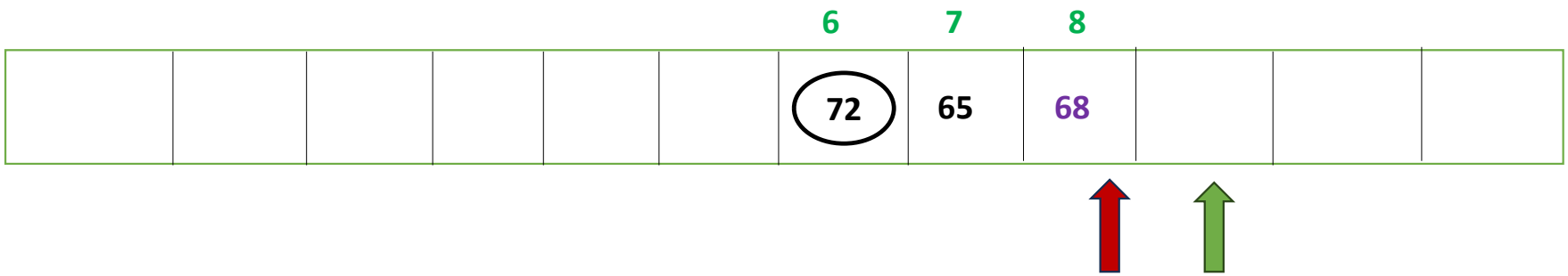

```
public static int partition(int arr[], int low, int high)
{                                                  6       8
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 72
  left=low;            // left = 6
  right =high;         // right = 8

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;
```



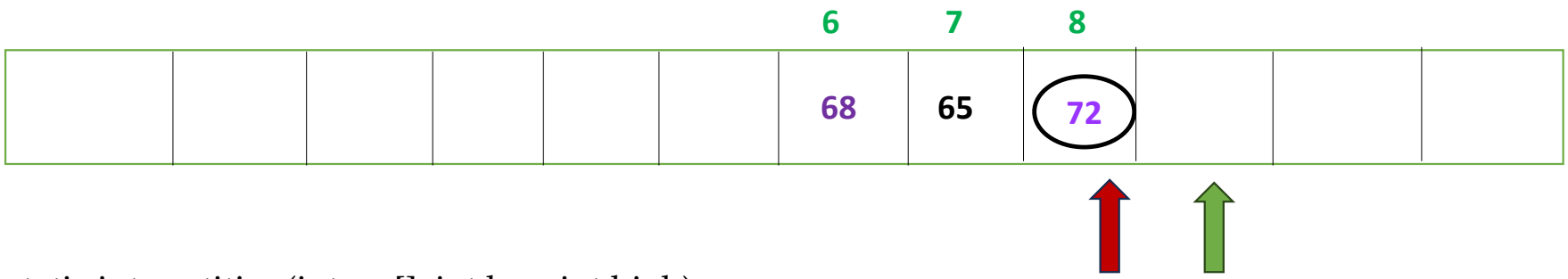

```
public static int partition(int arr[], int low, int high)
{                                                  6       8
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 72
  left=low;            // left = 6
  right =high;         // right = 8

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;
```




```
public static int partition(int arr[], int low, int high)
{                                                  6        8
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 72
  left=low;            // left = 6
  right =high;         // right = 8

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;

   if(left < right)    //false
      {
        t=arr[left];
        arr[left] = arr[right] ;
        arr[right] = t;
        left++; right--;
      }
      else left++;
   }  // end of while loop

   arr[low] = arr[right];
   arr[right] =pivot;

   return right;      // right =
}// end of function
```

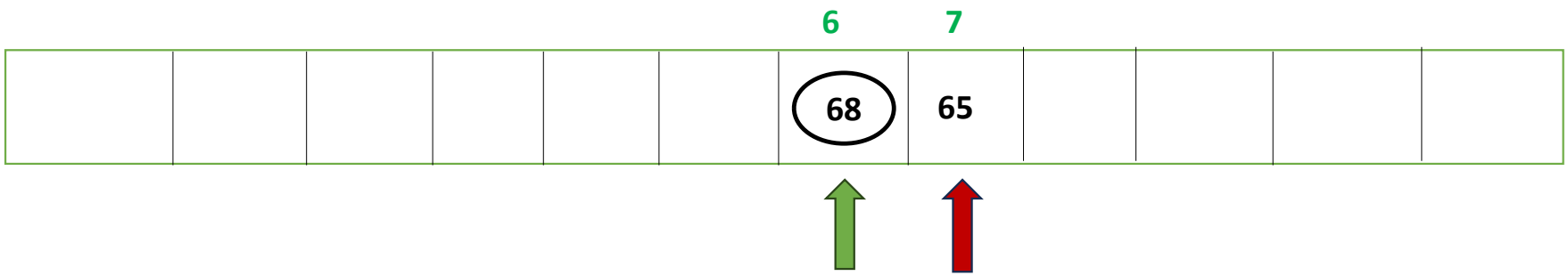



```
public static int partition(int arr[], int low, int high)
{                                                  6        8
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 72
  left=low;            // left = 6
  right =high;         // right = 8

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;

   if(left < right)    //false
   {
     t=arr[left];
     arr[left] = arr[right] ;
     arr[right] = t;
     left++; right--;
   }
   else left++;
  }  // end of while loop

  arr[low] = arr[right];
  arr[right] =pivot;

  return right;      // right =
}// end of function
```

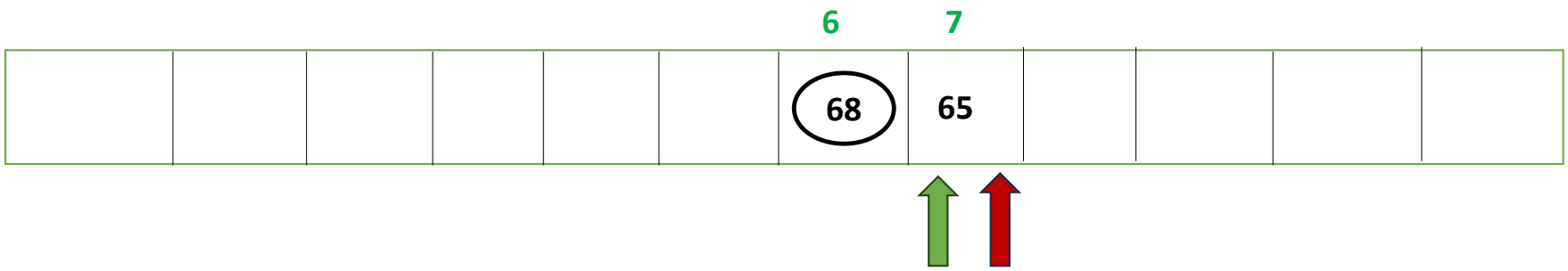



```
public static int partition(int arr[], int low, int high)
{                                          6       8
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 72
  left=low;            // left = 6
  right =high;         // right = 8

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;

   if(left < right)    //false
     {
       t=arr[left];
       arr[left] = arr[right] ;
       arr[right] = t;
       left++; right--;
     }
     else left++;
   }  // end of while loop

  arr[low] = arr[right];
  arr[right] =pivot;

  return right;  // right = 8
}// end of function
```

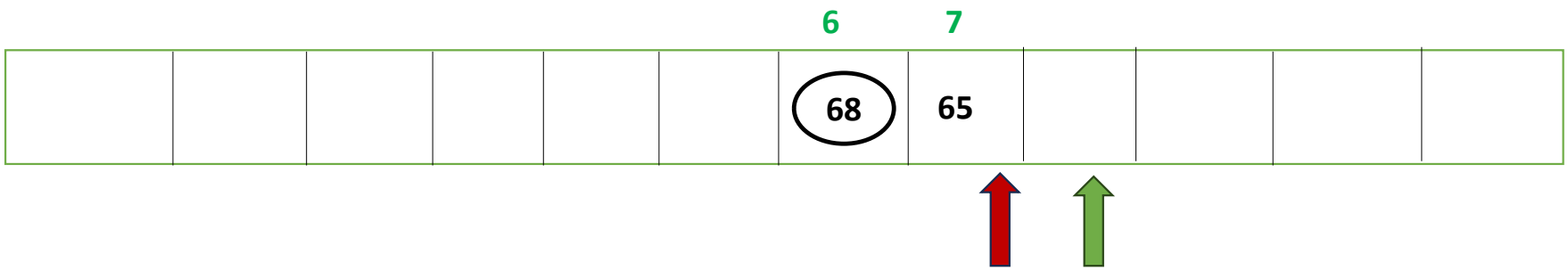
```
public static void QuickSort(int arr[],
              int low,int high)
{                  // 6        8
  int pivloc;

  if(low>=high)  return;

  pivloc = partition(arr,low,high); //8

  QuickSort(arr,low,pivloc-1);
                 //    6        7
  QuickSort(arr,pivloc+1, high);
                 //       9        8
}
```



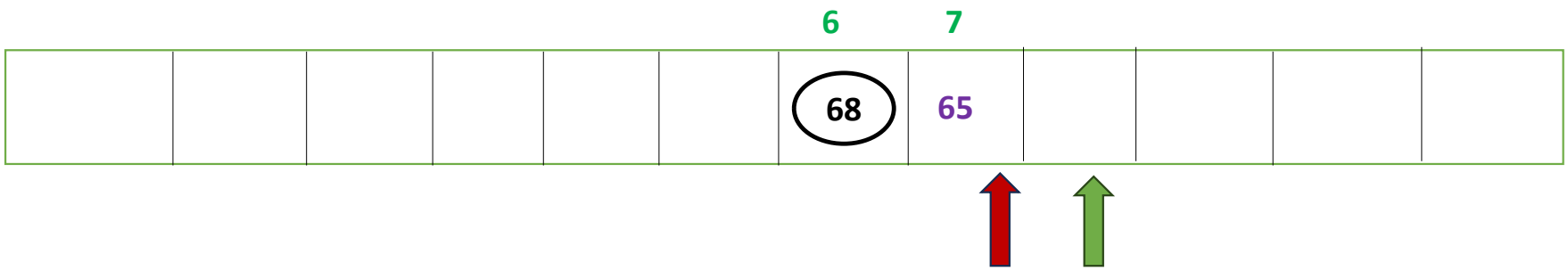

```
public static int partition(int arr[], int low, int high)
{                                                  6       7
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 68
  left=low;            // left = 6
  right =high;         // right = 7

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;
```




```
public static int partition(int arr[], int low, int high)
{                                                  6        7
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 68
  left=low;            // left = 6
  right =high;         // right = 7

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;
```




```
public static int partition(int arr[], int low, int high)
{                                                  6       7
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 68
  left=low;            // left = 6
  right =high;         // right = 7

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;

   if(left < right)    //false
     {
       t=arr[left];
       arr[left] = arr[right] ;
       arr[right] = t;
       left++; right--;
     }
     else left++;
  }  // end of while loop

  arr[low] = arr[right];
  arr[right] =pivot;

  return right;      // right = 8
}// end of function
```

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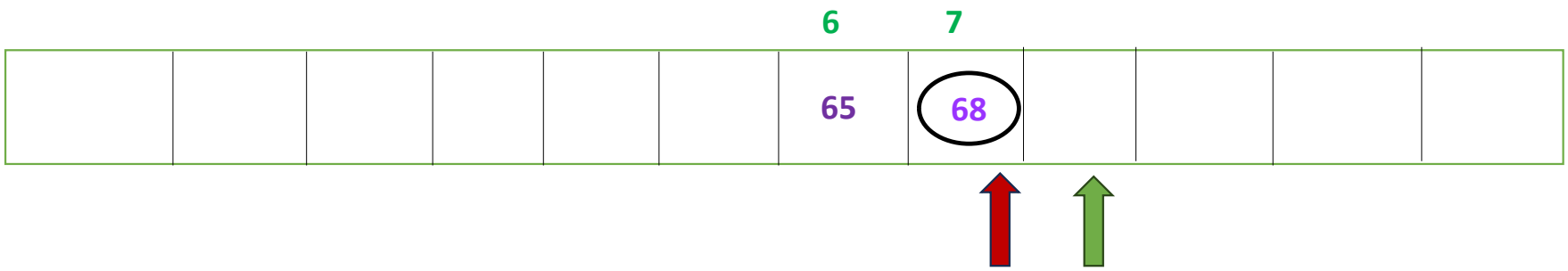
   if(left < right)    //false
   {
     t=arr[left];
     arr[left] = arr[right] ;
     arr[right] = t;
     left++; right--;
   }
   else left++;
  }  // end of while loop

  arr[low] = arr[right];
  arr[right] =pivot;

  return right;      // right =
8
}// end of function
```

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      left++;

   while(arr[right] > pivot)
      right--;

   if(left < right)    //false
   {
     t=arr[left];
     arr[left] = arr[right] ;
     arr[right] = t;
     left++; right--;
   }
   else left++;
  }  // end of while loop

  arr[low] = arr[right];
  arr[right] =pivot;

  return right;      // right =
7
}// end of function
```

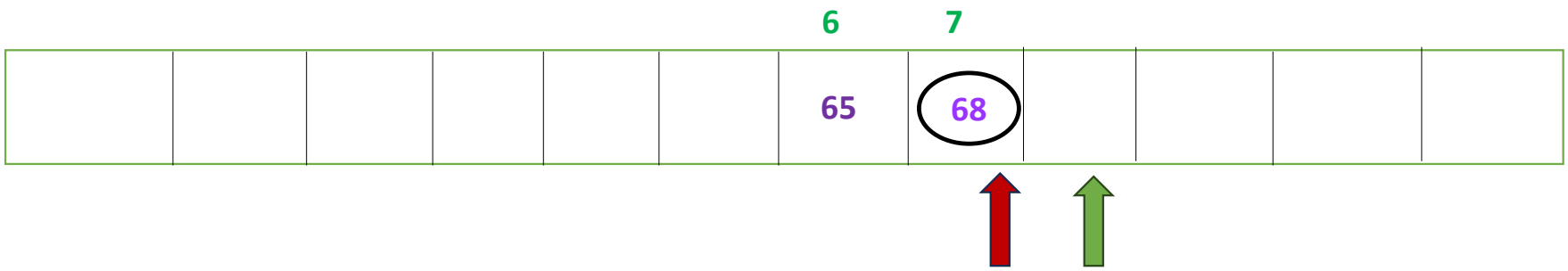


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   {
     t=arr[left];
     arr[left] = arr[right] ;
     arr[right] = t;
     left++; right--;
   }
   else left++;
  }  // end of while loop

  arr[low] = arr[right];
  arr[right] =pivot;

  return right;  // right = 7
}// end of function
```

```
public static void QuickSort(int arr[],
            int low,int high)
{                  // 6        7
  int pivloc;

  if(low>=high)  return;

  pivloc = partition(arr,low,high); //7

  QuickSort(arr,low,pivloc-1);
                 //     6         6
  QuickSort(arr,pivloc+1, high);
                 //       8         7
}
```

```
public static int partition(int arr[], int low, int high)
{                                                  5        11
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 95
  left=low;            // left = 5
  right =high;         // right = 11

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;

   if(left < right)    //false
      {
        t=arr[left];
        arr[left] = arr[right] ;
        arr[right] = t;
        left++; right--;
      }
      else left++;
   }  // end of while loop

   arr[low] = arr[right];
   arr[right] =pivot;

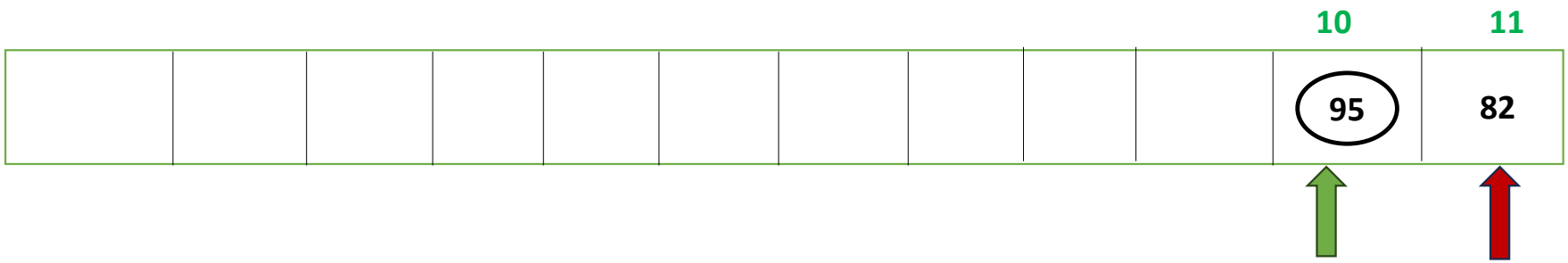
   return right; // right = 9
}// end of function
```

```
public static void QuickSort(int arr[],
              int low, int high)
{              // 5          11
  int pivloc;

  if(low>=high)  return;

  pivloc = partition(arr,low,high); //9

  QuickSort(arr,low,pivloc-1);
              //    5        8
  QuickSort(arr,pivloc+1, high);
              //    10       11
}
```

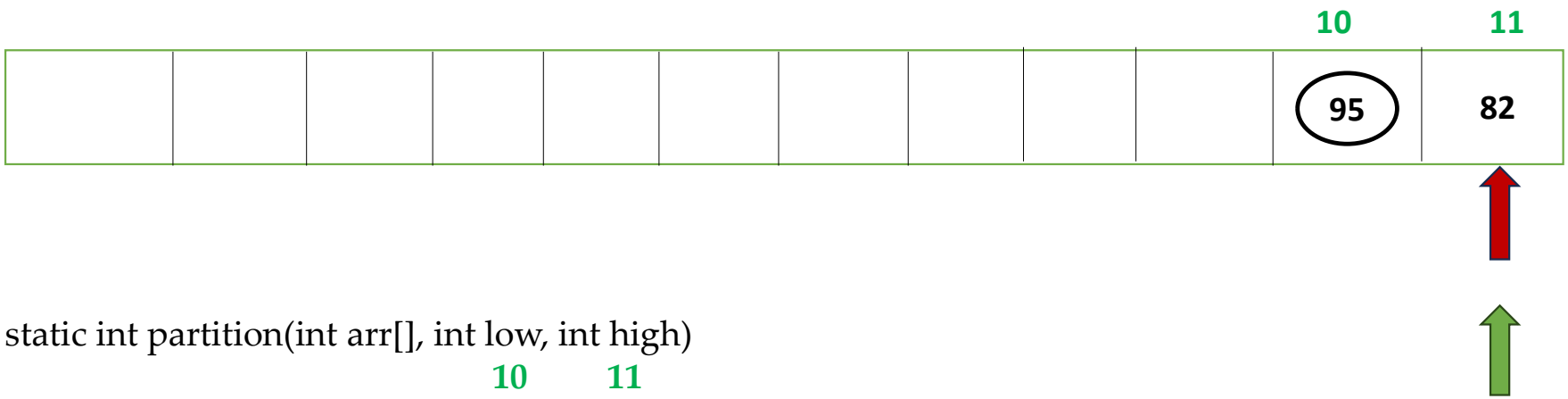



```
public static int partition(int arr[], int low, int high)
{                                                  10        11
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 95
  left=low;            // left = 10
  right =high;         // right = 11

  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;
```

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{                                                  10        11
  int left, right, pivot, t;

  pivot=arr[low];      // pivot = 95
  left=low;            // left = 10
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  while(left <= right)
  {
   while((arr[left]<=pivot) && (left<high))
      left++;

   while(arr[right] > pivot)
      right--;
```

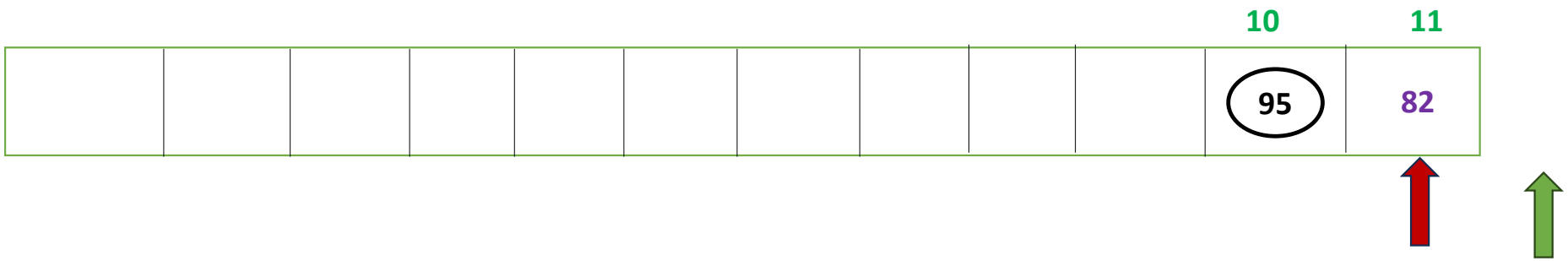


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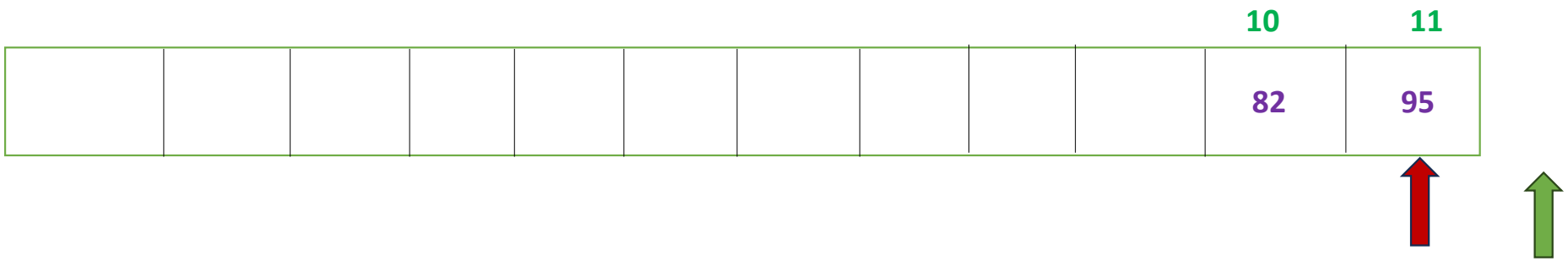
     {
       t=arr[left];
       arr[left] = arr[right] ;
       arr[right] = t;
       left++; right--;
     }
     else left++;
  }  // end of while loop

  arr[low] = arr[right];
  arr[right] =pivot;

  return right;      // right =
9
}// end of function
```




```
public static int partition(int arr[], int low, int high)
{                                                  10      11
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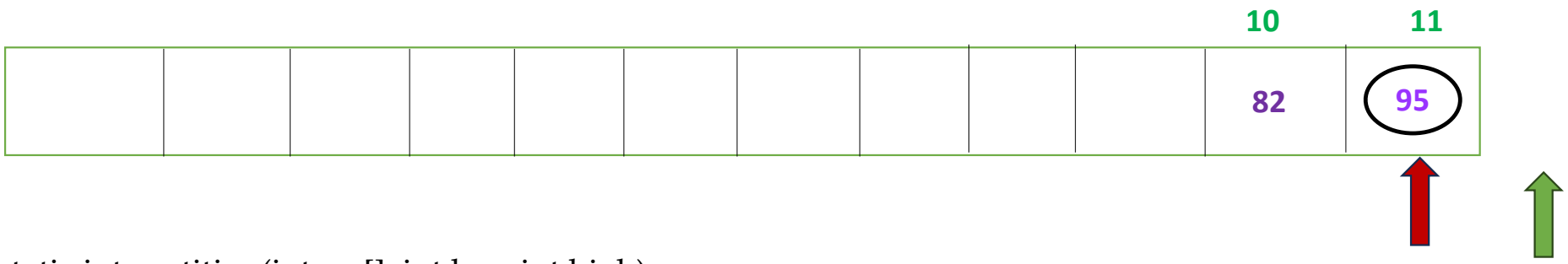
       arr[left] = arr[right] ;
       arr[right] = t;
       left++; right--;
     }
     else left++;
  }  // end of while loop

  arr[low] = arr[right];
  arr[right] =pivot;

  return right;      // right =
11
}// end of function
```

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       arr[left] = arr[right] ;
       arr[right] = t;
       left++; right--;
     }
     else left++;
   }  // end of while loop

   arr[low] = arr[right];
   arr[right] =pivot;

   return right;      // right = 11
}// end of function
```

```
public static void QuickSort(int arr[],
              int low, int high)
{              // 10          11
  int pivloc;

  if(low>=high)  return;

  pivloc = partition(arr,low,high); //11

  QuickSort(arr,low,pivloc-1);
              //     10        10
  QuickSort(arr,pivloc+1, high);
              //      12        11
}
```

Performance – Quick Sort

- Worst case Complexity: $O(n^2)$
- Best case Complexity: O(nlogn)
- Average case Complexity: O(nlogn)
- Worst case space Complexity: O(1)